Evaluating the Therapeutic Quality of LLMs in Mental Health Contexts

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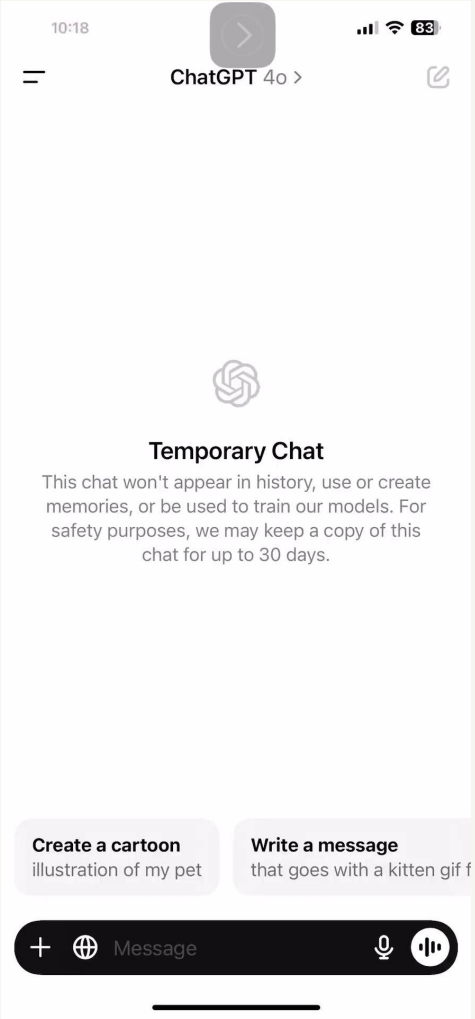
ChatGPT


10:18
83
ChatGPT 4o
Temporary Chat
This chat won't appear in history, use or create memories, or be used to train our models. For safety purposes, we may keep a copy of this chat for up to 30 days.
Create a cartoon
illustration of my pet
Write a message
that goes with a kitten gif f
Message

Two Goals:

1. Evaluate the therapeutic quality of LLMs for mental health

2. Fine-Tuning Llama for Improvements

Baseline Methodology

- Dataset
  - Amod/Mental Health Counseling Conversations
  - mpingale/Mental Health Chat Dataset
  - heliosbrahma/Mental Health Chatbot Dataset
- Compare Similarity of Models to Human Baseline Response
  - Word Based Metrics: BLEU, ROGUE
  - Semantic Similarity: BertScore, SentenceTransformers

Baseline Methodology

Context	Response
I've gone to a couple therapy sessions so far and still everytime I walk in I get nervous and shaky. Is this normal? Should I still be feeling like this?	Thank you for your question. It's completely normal and natural to feel nervous before a therapy session. Many people report having felt...

Baseline Methodology

- Prompt Selection
  - No Prompt
  - Prompt 1 - Minimal Guidance
    - *You are a licensed therapist. Your role is to respond to the user's question with empathy, understanding, and support.*
  - Prompt 2 - Empathetic Guidance
    - *You are a licensed therapist. Your role is to respond to the user's question with empathy, understanding, and support. Empathy means recognizing and validating the user's feelings while offering thoughtful, non-judgmental responses*

Baseline Models - Results

Model	Prompts	BLEU	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore	Sentence Transformer (all-mpnet-base-v2)
Therapist Baseline		0.4362	0.5376	0.3680	0.4343	Precision: 0.6871, Recall: 0.6951, F1: 0.6881	N/A
LLaMA 3.1 8B Instruct	No Prompt	0.0113	0.2167	0.0366	0.1065	Precision: 0.4984, Recall: 0.5549, F1: 0.5234	0.6331
	Prompt 1	0.0134	0.2581	0.0380	0.1205	Precision: 0.4955, Recall: 0.5263, F1: 0.5079	0.6088
	Prompt 2	0.0139	0.2675	0.0388	0.1251	Precision: 0.5113, Recall: 0.5355, F1: 0.5208	0.6204
FLAN-T5	No Prompt	0.0000	0.0591	0.0079	0.0430	Precision: 0.4749, Recall: 0.2997, F1: 0.3641	0.2779
	Prompt 1	0.0023	0.1273	0.0179	0.0923	Precision: 0.5367, Recall: 0.3926, F1: 0.4510	0.4202
	Prompt 2	0.0021	0.1300	0.0208	0.0955	Precision: 0.5414, Recall: 0.3902, F1: 0.4512	0.4254
T5	No Prompt	0.0015	0.0914	0.0090	0.0673	Precision: 0.4556 Recall: 0.3703 F1: 0.4064	0.3574

- Good performance on LLaMA
- How can we do better?

Fine-Tuning Llama

Fine-Tuning Setup

- Llama 3.1 8B Instruct
- qLoRA
- Data

Context	Response
I've gone to a couple therapy sessions so far and still everytime I walk in I get nervous and shaky. Is this normal? Should I still be feeling like this?	Thank you for your question. It's completely normal and natural to feel nervous before a therapy session. Many people report having felt...

Fine-Tuning Experiments (Part 1)

input_ids	labels
You are a licensed therapist. Your role is to respond to the user's question with empathy, understanding, and support. Empathy means recognizing and validating the user's feelings while offering thoughtful, non-judgmental responses. User: **I've gone to a couple therapy sessions so far and still everytime I walk in I get nervous and shaky. Is this normal? Should I still be feeling like this?** Therapist:	**Thank you for your question. It's completely normal and natural to feel nervous before a therapy session. Many people report having felt...**

Small Training (525 examples)

==========It’s completely normal to feel nervous or shaky when attending therapy sessions, especially if you’re new to the process. Many people experience anxiety…

Full Training (4719 examples)

with, that a the for is you and a to to what a you you, a you feel or to the to with. in in the you. to you,

Problem & Fix

input_ids	labels
You are a licensed therapist. Your role is to respond to the user's question with empathy, understanding, and support. Empathy means recognizing and validating the user's feelings while offering thoughtful, non-judgmental responses. User: **I've gone to a couple therapy sessions so far and still everytime I walk in I get nervous and shaky. Is this normal? Should I still be feeling like this?** Therapist: **Thank you for your question. It's completely normal and natural to feel nervous before a therapy session. Many people report having felt...**	You are a licensed therapist. Your role is to respond to the user's question with empathy, understanding, and support. Empathy means recognizing and validating the user's feelings while offering thoughtful, non-judgmental responses. User: **I've gone to a couple therapy sessions so far and still everytime I walk in I get nervous and shaky. Is this normal? Should I still be feeling like this?** Therapist: **Thank you for your question. It's completely normal and natural to feel nervous before a therapy session. Many people report having felt...**

Remaining Issue

Small Training (525 examples)

3% no response

Full Training (4719 examples)

70% no response 😱

Fine-Tuning Experiments (Part 2)

- Extracted Logits to Verify Probability Distributions
- Fixed Decoding Bug
- Achieved Consistent Responses 😃

Evaluation Results and Discussion

Metric	LLaMA Baseline	LLaMA Fine-Tuned
BLEU	0.0139	0.0111
ROUGE-1	0.2675	0.2436
ROUGE-2	0.0388	0.0333
ROUGE-L	0.1251	0.1267
BERTScore (F1)	0.5208	0.5076

An Example of the Fine-Tuned Model Output:

Hello, and thanks for your question. It's normal to feel nervous or anxious when you are doing something new or uncomfortable. In therapy, you are being vulnerable and sharing your thoughts and feelings with someone new. It's normal to feel nervous and it will take some time to get used to it. It might help you to know that the therapist is trained to be a safe place for you to talk and share your thoughts and feelings. If you are feeling nervous or anxious, it might be helpful to let your therapist know. They can help you work through your feelings and find ways to make you feel more comfortable. It's also okay if you don't feel like you want to go back to therapy. It's okay to try something new and then decide it's not for you. But, if you do decide to go back to therapy, you might want to talk to your therapist about how you are feeling. They can help you work through your feelings and find ways to make you feel more comfortable. Remember, therapy is a safe place for you to talk and share your thoughts and feelings. It's okay to feel nervous, but it's not okay to let your nervousness stop you from trying something that might be helpful to you. It might help you to know that therapy is not a one size fits all…

Thank You!